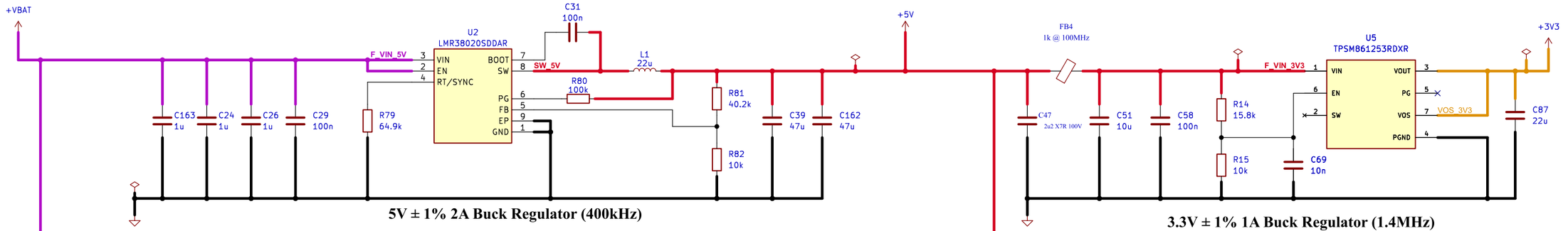
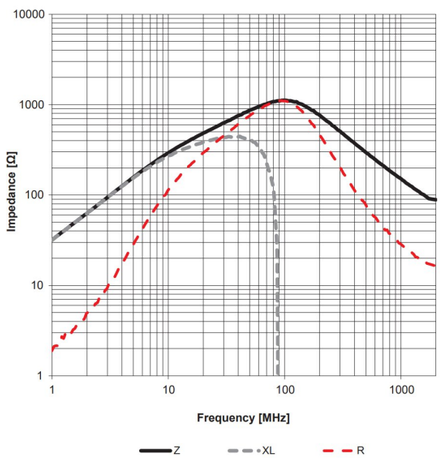
[6] Power - Generation



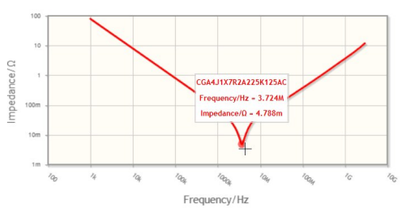
Voltage Regulators

Ferrite bead Impedance vs Frequency curve



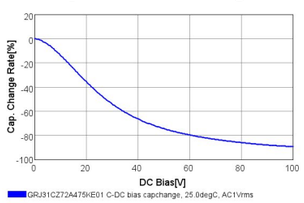
DESIGN NOTE:
XL drops around 45th harmonic.

Pi filter capacitor Impedance vs Frequency curve



CGA4J1X7R2A225K125AC

Bulk capacitance Derating curve



DESIGN NOTE:
Effective capacitance at 33V is approximately 2.1 uF

Capacitor and Ferrite Bead Parameters

Battery

+VBAT	TP39	+VBAT
+VBAT	TP40	+VBAT
+VBAT	TP41	+VBAT

Filtered Power Rails

F_VIN_12V	TP42	F_VIN_12V
F_VIN_5V	TP43	F_VIN_5V
F_VIN_3V3	TP44	F_VIN_3V3

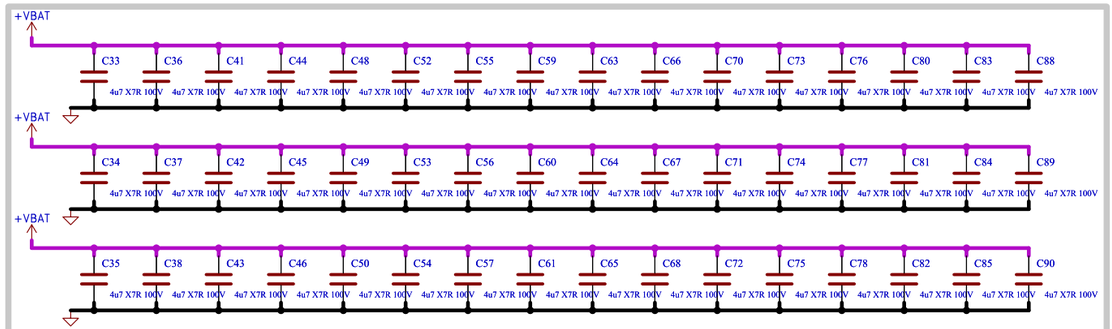
Power Rails

+12V	TP45	+12V
+5V	TP46	+5V
+5V	TP47	+5V
+3V3	TP48	+3V3
+3V3	TP49	+3V3
+3V3	TP50	+3V3
+A3V3	TP51	+A3V3

Ground

GND	TP52	GND
GND	TP53	GND
GND	TP54	GND
GND	TP55	GND
GND	TP56	GND
GND	TP57	GND
GND	TP58	GND

Test Points



DESIGN NOTE:
An external precharge circuit is advised to charge the bulk capacitors if multiple boards are connected.

DC-Link Bulk Capacitors. 48 x 4.7 uF = 225.6 uF (100.8 uF at 33V)

DESIGN NOTE:
X6S 10uF is marginally better (but worse DC bias derating) at a higher price.

Comments: (+12V (600 mA): Fan power) +5V (600 mA): Gate Driver, LEDs +3V3 (1 A): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: **ETH Robotics Club**		Variant: PRELIMINARY	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: **A3**	Sheet: **6** of **20**